



PsyPredict v2.0 — Complete Presentation Guide

Prepared for: Saphalya Date: March 2026 Project: PsyPredict — Production-Grade Multimodal Clinical AI System



PART 1 — PPT PRESENTATION GUIDE

*Use this section as a **slide-by-slide script** for your PowerPoint / Google Slides presentation.*

SLIDE 1 — Title Slide

Title: *PsyPredict v2.0 — AI-Powered Multimodal Mental Health Analysis*

Subtitle: A fully local, privacy-first clinical AI system for real-time emotional assessment

Visuals: Project logo / clean dark gradient background with neural-network mesh art

Talking Points:

- Introduce yourself: name, role, institution
 - One-liner: "PsyPredict uses four AI models working together to understand how someone is feeling — from their words, their face, and the clinical danger signals in between."
-

SLIDE 2 — The Problem

Title: *The Mental Health Gap*

Key Stats to Display:

- 1 in 8 people globally live with a mental disorder (WHO, 2022)
- India has < 1 psychiatrist per 100,000 people
- 70%+ of mental health conditions go undiagnosed
- Crisis detection is still keyword-based in most apps — dangerously unreliable

Talking Points:

- Mental health screening is slow, inaccessible, and often missed
 - Most mental health apps use basic sentiment analysis — no clinical depth
 - Crisis detection based on keywords like "suicide" misses nuanced distress signals
 - There is a **massive gap** between what AI can do and what mental health tools actually do today
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SLIDE 3 — Our Solution

Title: *PsyPredict — What It Does*

Bullet Points:

1. **Multimodal Analysis** — Understands emotions from text AND facial expressions simultaneously
2. **Clinical-Grade Reports** — Every response returns a structured `PsychReport`, not a chatbot reply
3. **Crisis Safety Net** — Zero-shot AI crisis detection that overrides the system when danger is detected
4. **Fully Local** — All AI runs on your machine. No data leaves. No cloud APIs.

5. **Culturally Grounded** — Includes Bhagavad Gita–based remedies for holistic guidance

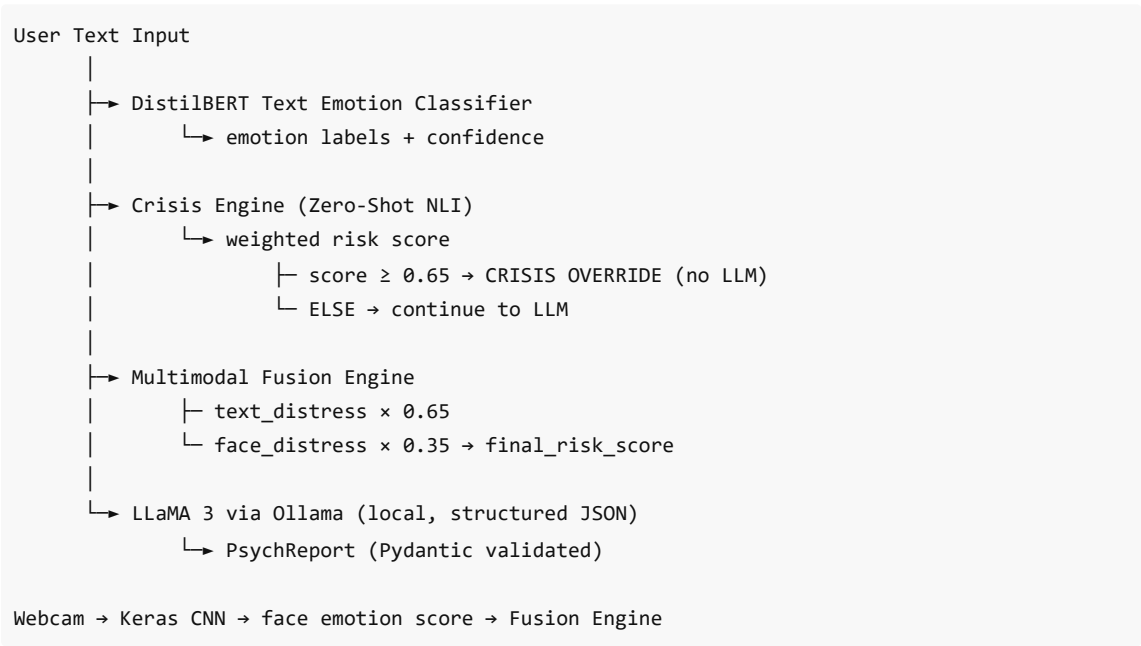
Talking Points:

- This is NOT a chatbot. It's a clinical decision-support system that produces structured psychological assessments
- Every single response is validated through Pydantic schemas — no hallucinated junk
- The crisis layer runs BEFORE the LLM — if someone is in danger, the system responds immediately with helpline numbers

SLIDE 4 — System Architecture

Title: *How It Works — Architecture Overview*

Visual: Display the architecture diagram (use `DOCUMENTATION/PIPELINE.PNG` or recreate)



Talking Points:

- Walk through the flow top-to-bottom
- Emphasize: "The crisis engine runs BEFORE the LLM — if it triggers, the LLM is never called"
- Explain the fusion formula: $(\text{Text} \times 0.65) + (\text{Face} \times 0.35) = \text{Final Risk Score}$
- Highlight Pydantic validation: every output is Schema-enforced, never free-form text

SLIDE 5 — The AI Models

Title: *Four AI Engines Working Together*

#	Engine	Model	Purpose
1	Text Emotion	DistilBERT (bhadresh-savani/distilbert-base-uncased-emotion)	Multi-label emotion classification from user text

2	Crisis Detection	MiniLM Zero-Shot NLI (cross-encoder/nli-MiniLM2-L6-H768)	Detects suicidal ideation, self-harm across 5 risk dimensions
3	Face Emotion	Custom Keras CNN + OpenCV Haar Cascade	7-class real-time facial emotion detection via webcam
4	Clinical Reasoning	LLaMA 3 via Ollama (fully local)	Generates structured PsychReport with clinical-grade reasoning

Talking Points:

- DistilBERT: lightweight transformer, 6 emotion classes — sadness, joy, fear, anger, surprise, love
- MiniLM NLI: this is NOT keyword matching — it uses natural language inference to understand MEANING
- Keras CNN: trained on FER2013 dataset, 7 classes (happy, sad, angry, fear, disgust, surprise, neutral)
- LLaMA 3: runs entirely locally via Ollama — zero data leaves the machine

SLIDE 6 — PsychReport Output

Title: *Structured Clinical Output — The PsychReport*

Display a sample JSON:

```
{
  "risk_classification": "MODERATE",
  "emotional_state_summary": "Elevated anxiety with undertones of helplessness.",
  "behavioral_inference": "User shows avoidance patterns and withdrawal.",
  "cognitive_distortions": ["Catastrophizing", "Mind Reading"],
  "suggested_interventions": [
    "Practice grounding exercises (5-4-3-2-1 technique)",
    "Schedule a session with a licensed therapist"
  ],
  "confidence_score": 0.78,
  "crisis_triggered": false
}
```

Talking Points:

- Every response is structured — not free-form chatbot text
- Risk levels: MINIMAL → LOW → MODERATE → HIGH → CRITICAL
- Cognitive distortions are CBT-based labels (catastrophizing, black-and-white thinking, etc.)
- The confidence score is the LLM's own self-assessed reliability metric
- If `crisis_triggered = true`, the report includes emergency resources automatically

SLIDE 7 — Crisis Detection Deep-Dive

Title: *The Safety Net — Zero-Shot Crisis Detection*

Key Points:

- Uses **Natural Language Inference (NLI)** — not keyword matching
- Evaluates across 5 weighted risk dimensions:

Risk Dimension	Weight
Suicidal ideation	1.00
Self-harm intent	1.00
Immediate danger to self	0.95
Severe mental breakdown	0.60
Hopelessness and worthlessness	0.50

- Threshold: ≥ 0.65 → **CRISIS OVERRIDE** → LLM is bypassed entirely
- Emergency resources (iCall, Vandrevalla, AASRA) served automatically

Talking Points:

- "The phrase '*I don't see a future for myself*' would NOT be caught by keyword matching. Our NLI model understands its meaning."
- This layer runs BEFORE the LLM — it's the safety net of the entire system
- The response is deterministic — not generated by AI — to ensure reliability in critical moments

SLIDE 8 — Tech Stack

Title: *Technology Stack*

Layer	Technology
Frontend	React 18, Vite, TypeScript, TailwindCSS
Backend	Python 3.10, FastAPI, Uvicorn, Pydantic v2
LLM	LLaMA 3 via Ollama (fully local, no API keys)
Text Emotion	HuggingFace Transformers — DistilBERT
Crisis NLI	HuggingFace Transformers — MiniLM Zero-Shot
Face Emotion	OpenCV + Custom Keras/TensorFlow CNN
Database	Supabase (PostgreSQL + Auth + RLS)
Knowledge Base	Pandas + CSV (Bhagavad Gita remedies)
Deployment	Docker Compose, Vercel (frontend)
API Docs	Swagger UI (auto-generated)

Talking Points:

- Everything runs locally — no OpenAI, no Google API, no cloud LLM
 - Supabase provides user authentication + row-level security — each user can only see their own data
 - Docker Compose for one-command deployment of the entire stack
-

SLIDE 9 — Live Demo (if applicable)

Title: *Live Demonstration*

Demo Flow:

- 1. Open the app → show the landing page
- 2. Type a message expressing mild anxiety → show the PsychReport panel
- 3. Toggle webcam → show real-time facial emotion in the sidebar
- 4. Type a high-risk message → show the crisis override in action
- 5. Navigate to history → show conversation storage with Supabase

Tips:

- Have Ollama running beforehand (`ollama serve` in a terminal)
- Have the backend and frontend already started
- Pre-test the crisis trigger with: *"I don't want to be here anymore, nothing matters"*

SLIDE 10 — Future Roadmap

Title: *What's Next*

Phase	Feature
v2.1	Speech emotion analysis (audio modality)
v2.2	TAT (Thematic Apperception Test) narrative analysis engine
v2.3	Session-over-session longitudinal tracking
v3.0	NeuroLink integration — EEG/EMG neural signal decoding
v3.x	Multi-language support (Hindi, Tamil, Bengali)
v4.0	Institutional deployment (universities, corporate wellness programs)

Talking Points:

- The fusion engine already has a `speech_score` placeholder — designed for expansion
- TAT analysis is already built separately and can be integrated
- NeuroLink is a sister project for neural-signal-based communication

SLIDE 11 — Thank You + Q&A

Content:

- "Thank you" with team/contact info
 - GitHub repo link
 - Live demo URL: `psypredict.vercel.app`
 - Disclaimer: *"PsyPredict is a clinical decision-support tool for educational purposes. It does not replace professional therapy."*
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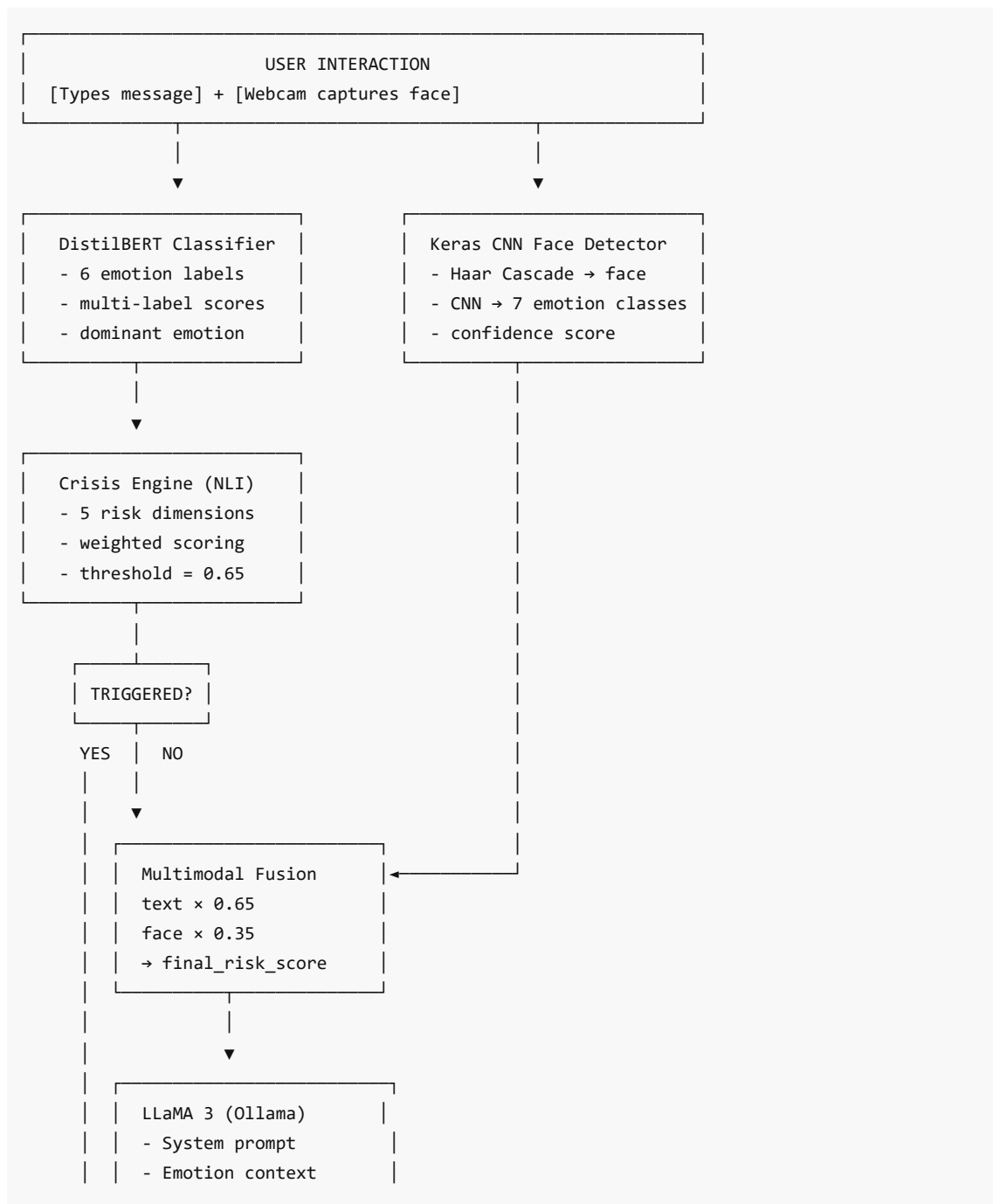
PART 2 — WORKFLOW, TECHNICAL DEEP-DIVE & BUSINESS PITCH

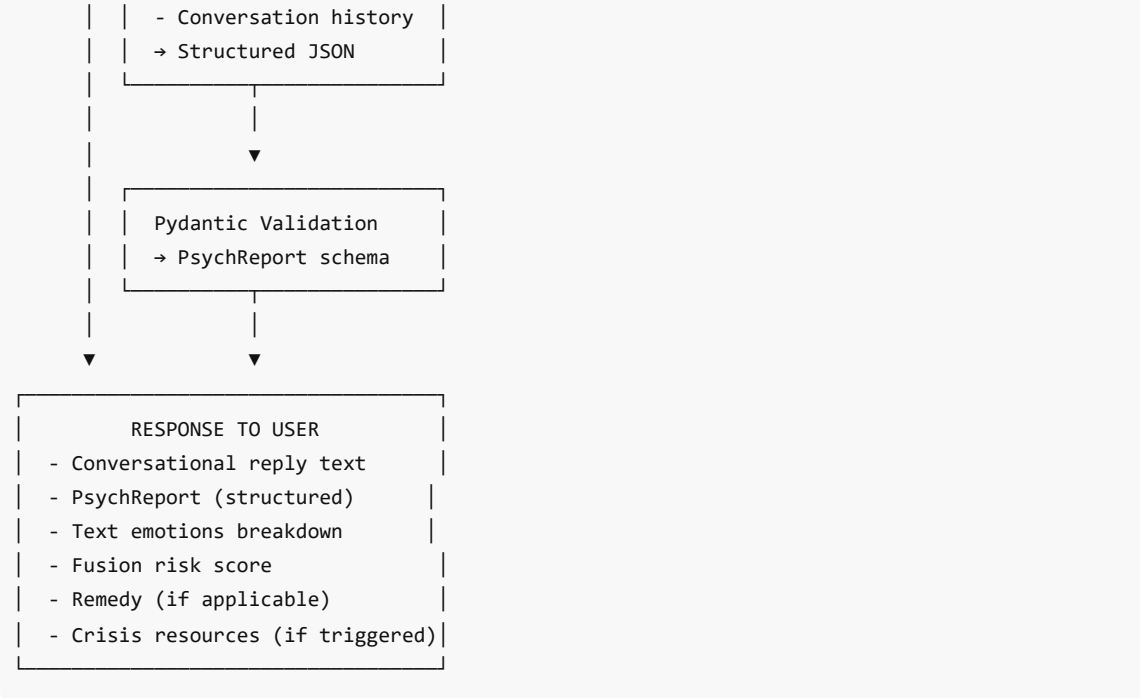
Use this section for *viva voce*, investor pitches, hackathon judging, or detailed technical Q&A.



I. System Workflow — Step by Step

Complete Request Lifecycle





Key Workflow Principles

- 1. **Crisis-First Architecture** — Safety layer evaluates BEFORE the LLM is invoked
- 2. **Multimodal Fusion** — Never relies on a single source; combines text + visual signals
- 3. **Schema-Enforced Outputs** — Pydantic validation rejects malformed LLM responses
- 4. **Graceful Degradation** — If Ollama is down, a fallback report is returned with `service_degraded: true`
- 5. **Context Windowing** — Only last 10 conversation turns are sent to the LLM to prevent token overflow

II. Architecture Types & Patterns

Architecture Classification

Dimension	Classification
Overall	Modular Monolith (backend) + SPA (frontend)
API Pattern	RESTful JSON API (FastAPI + auto-docs)
AI Pattern	Pipeline Architecture (sequential engine chain)
Data Pattern	Event-driven per-request (no batch processing)
Auth Pattern	Supabase Auth + Row-Level Security (RLS)
Deployment	Containerized (Docker Compose)

Design Patterns Used

Pattern	Where It's Used
Singleton	<code>crisis_engine</code> , <code>fusion_engine</code> , <code>text_emotion_engine</code>

Strategy	Emotion-to-distress score mapping (configurable)
Pipeline	Sequential engine chain: Text → Crisis → Fusion → LLM → Validation
Fallback/Circuit Breaker	Ollama retry with exponential backoff + fallback report
Dependency Injection	Settings loaded via <code>get_settings()</code> → engines read from config
Schema Validation	All I/O enforced through Pydantic models

III. Models — Detailed Breakdown

Model 1: DistilBERT Text Emotion Classifier

Attribute	Detail
Model	bhadresh-savani/distilbert-base-uncased-emotion
Type	Transformer (distilled BERT)
Task	Multi-label text emotion classification
Classes	sadness, joy, fear, anger, surprise, love
Input	User text (max 512 tokens)
Output	List of {label, score} pairs
Why This Model	Lightweight (66M params), fast inference on CPU, well-validated on emotion datasets

Model 2: MiniLM Zero-Shot NLI (Crisis Detection)

Attribute	Detail
Model	cross-encoder/nli-MiniLM2-L6-H768
Type	Cross-encoder Transformer (NLI)
Task	Zero-shot classification across 5 risk dimensions
Input	User text (max 512 chars) + candidate risk labels
Output	Per-label confidence scores → weighted sum → risk score
Why This Model	No fine-tuning needed; understands meaning, not just keywords; fast on CPU

Model 3: Custom Keras CNN (Facial Emotion)

Attribute	Detail
Model	Custom CNN trained on FER2013
Type	Convolutional Neural Network
Task	7-class facial emotion classification

Classes	happy, sad, angry, fear, disgust, surprise, neutral
Input	48×48 grayscale face crop (via Haar Cascade)
Output	Predicted class + confidence score
Why This Model	Lightweight, runs in real-time on webcam feed, no GPU required

Model 4: LLaMA 3 (Clinical Reasoning)

Attribute	Detail
Model	LLaMA 3 8B via Ollama
Type	Large Language Model (decoder-only Transformer)
Task	Structured clinical reasoning → PsychReport JSON
Input	System prompt + emotion context + conversation history
Output	JSON conforming to PsychReport schema
Why This Model	Fully local (no API keys), strong instruction-following, JSON-mode support

Fusion Formula

```
text_distress = map(dominant_text_emotion → distress_score)
face_distress = map(face_emotion → distress_score)

final_risk_score = (TEXT_WEIGHT × text_distress) + (FACE_WEIGHT × face_distress)
                  = (0.65 × text_distress) + (0.35 × face_distress)
```

Distress score mappings:

- **Text:** sadness=0.85, fear=0.80, anger=0.60, disgust=0.50, surprise=0.30, joy=0.05
- **Face:** fear=0.80, sad=0.70, angry=0.50, disgust=0.40, surprised=0.30, neutral=0.20, happy=0.05

 IV. Business Pitch

 The Problem (Market Gap)

- **970 million** people worldwide suffer from mental disorders (WHO)
- India: **150 million+** people need mental health support; **< 9,000** psychiatrists available
- Existing apps are glorified chatbots — no clinical depth, no multimodal analysis, no real crisis detection
- Most solutions require cloud APIs → **privacy nightmare** for sensitive mental health data

 Our Solution

PsyPredict is the **first fully-local, multimodal, clinical-grade mental health AI system** that:

1. Combines **4 AI models** for accurate emotional understanding
2. Produces **structured clinical reports**, not chatbot replies
3. Runs **100% locally** — zero data leaves the user's machine
4. Has a **deterministic crisis safety net** that cannot be fooled by paraphrasing

👥 Target Users

Segment	Use Case
University Counseling Centers	Triage incoming students, prioritize high-risk cases
Corporate Wellness Programs	Anonymous employee mental health screening
Telehealth Platforms	Pre-session assessment before therapist consultation
Clinical Psychologists	Decision-support tool during sessions
Individual Users	Self-awareness and emotional tracking
NGOs & Government Programs	Scalable mental health screening in underserved areas
Schools (K-12)	Early detection of at-risk students

★ Unique Selling Points (USPs)

#	USP	Why It Matters
1	Fully Local AI	No cloud, no data leaks — HIPAA/DISHA compliant by design
2	Multimodal Fusion	Text + Face = more accurate than single-modality systems
3	Structured Clinical Output	Every response is a validated PsychReport, not free-form text
4	NLI-Based Crisis Detection	Understands meaning, not keywords — catches subtle distress
5	LLM Override Safety Net	Crisis layer is deterministic — AI cannot hallucinate in emergencies
6	Culturally Grounded	Bhagavad Gita remedies for Indian user base
7	Open Source	MIT License — inspectable, auditable, trustworthy
8	Zero API Cost	No OpenAI/Google bills — runs on commodity hardware

📈 Scalability & Expansion Plan

PHASE 1 (Current)	PHASE 2 (6 months)	PHASE 3 (12 months)
- Text + Face - LLaMA 3 local - Single user - Docker local - English only	+ Speech Emotion + TAT Narrative Analysis - Multi-user (Supabase Auth) - Cloud deploy (GPU VMs) - Session analytics	+ EEG/EMG (NeuroLink) + Multi-language (Hindi, Tamil) - Institutional dashboards - White-label SaaS - Therapist-in-the-loop mode

Scaling Strategy:

- Horizontal:** Deploy on cloud VMs with GPU for institutional use
- Vertical:** Add more modalities (speech, neural signals)
- Geographic:** Multi-language support for Indian regional languages
- Market:** White-label SaaS for hospitals, universities, corporates
- Data:** Longitudinal tracking for treatment efficacy measurement

💰 Revenue Model (Potential)

Stream	Description
Freemium	Free for individuals, paid for advanced analytics
Institutional License	Annual SaaS license for universities/hospitals
API-as-a-Service	Sell the PsychReport pipeline to other mental health apps
White-Label	Custom-branded deployments for corporate clients
Research Grants	Collaborate with psychology departments for funded studies

🏆 Credibility & Validation

Factor	Evidence
Clinical Framework	Based on CBT (Cognitive Behavioral Therapy) distortion detection
Validated Models	DistilBERT trained on GoEmotions; FER2013 for facial emotion
Safety Engineering	Zero-shot NLI crisis detection with deterministic overrides
Production Quality	Input sanitization, rate limiting, exponential backoff, structured logging
Data Privacy	Fully local inference; Supabase RLS for user data isolation
Open Source	MIT License — full code transparency and auditability
Disclaimer	Clearly stated: decision-support tool, not a replacement for therapy

🔧 V. Technical Stack — Complete Reference

Backend Stack

Component	Technology	Version / Detail
Language	Python	3.10
Framework	FastAPI	Async, auto-docs
Server	Uvicorn	ASGI server
Validation	Pydantic v2	All I/O schemas
ML - Text	HuggingFace Transformers	DistilBERT, MiniLM
ML - Vision	TensorFlow/Keras, OpenCV	CNN + Haar Cascade
LLM	Ollama	LLaMA 3 (local)
Data	Pandas	CSV knowledge base

Frontend Stack

Component	Technology	Version / Detail
Language	TypeScript	Strict mode
Framework	React	18.x
Build Tool	Vite	Fast HMR
Styling	TailwindCSS	Utility-first
HTTP Client	Axios	Typed API calls
Routing	React Router	SPA navigation
State	React Context	Auth + chat state
Webcam	HTML5 MediaDevices API	Live feed

Infrastructure

Component	Technology
Database	Supabase (PostgreSQL)
Auth	Supabase Auth + RLS
Containerization	Docker + Docker Compose
Frontend Hosting	Vercel
Backend Hosting	Any VM with Ollama (GPU recommended)
API Documentation	Swagger UI (auto-generated at /docs)

Database Schema (Supabase)

```
conversations (id, user_id, created_at, title)
  |
  └─ messages (id, user_id, conversation_id, content, metadata, created_at)

profiles (id, updated_at, full_name, avatar_url, email, password_hash)
```

- Row-Level Security (RLS) enabled on all tables
- Users can only read/write their own data
- Profiles decoupled from `auth.users` to prevent race conditions

VI. Common Viva / Q&A Questions & Answers

Q1: "Why not just use ChatGPT?"

*ChatGPT is a cloud-based general chatbot. PsyPredict is a **local, multimodal clinical AI** that combines 4 specialized models, produces structured clinical reports, and has a deterministic crisis safety net. ChatGPT sends all data to OpenAI's servers — PsyPredict keeps everything on your machine.*

Q2: "How is your crisis detection better than keyword matching?"

We use Natural Language Inference (NLI) via MiniLM. The sentence "I don't see any reason to continue" has zero crisis keywords but our model scores it as high-risk because it understands the meaning. Keyword matchers would miss it entirely.

Q3: "What if the LLM hallucinates?"

Every LLM output is validated against a Pydantic schema (PsychReport). If the output doesn't conform, it's rejected. For critical cases, the crisis layer bypasses the LLM entirely — the response is deterministic and pre-written.

Q4: "Can this replace a therapist?"

No, and it's not designed to. PsyPredict is a **clinical decision-support tool** — it helps triage, screen, and support. It explicitly directs high-risk users to professional help. Our disclaimer makes this clear.

Q5: "Why run locally instead of using cloud APIs?"

Mental health data is among the most sensitive personal data. Sending it to third-party APIs creates privacy, compliance, and cost risks. Local inference via Ollama means **zero data exposure** and **zero API costs**.

Q6: "What is multimodal fusion and why does it matter?"

A person might type "I'm fine" while their face shows fear. Single-modality analysis would miss this. Our fusion engine combines text emotion (65% weight) and facial emotion (35% weight) to produce a more accurate risk score.

Q7: "How accurate is the system?"

DistilBERT achieves ~93% accuracy on the GoEmotions benchmark. The Keras CNN achieves ~65% on FER2013 (standard for the field). The fusion of multiple modalities compensates for individual model weaknesses.

Q8: "What makes this production-ready?"

Input sanitization (HTML stripping, char limits), rate limiting (30 req/min), exponential backoff on Ollama failures, graceful fallbacks, structured logging, Pydantic validation on all I/O, context window trimming, and Docker deployment.

 VII. API Reference (Quick)

Method	Endpoint	What It Does
POST	/api/chat	Full clinical pipeline → PsychReport + conversational reply
POST	/api/predict/emotion	Facial emotion detection from webcam frame
POST	/api/analyze/text	Text emotion analysis + crisis pre-screen
GET	/api/get_advice?condition=	Bhagavad Gita remedy lookup
GET	/api/health	System health check (Ollama + DistilBERT status)

 VIII. Presentation Checklist

Before the Presentation

- ☐ Ollama is installed and running (`ollama serve`)
- ☐ LLaMA 3 model is pulled (`ollama list` shows `llama3`)
- ☐ Backend is running (`uvicorn app.main:app --host 0.0.0.0 --port 7860`)
- ☐ Frontend is running (`npm run dev` → `localhost:5173`)
- ☐ Webcam is working and browser has camera permission
- ☐ Swagger UI is accessible at `localhost:7860/docs`
- ☐ Test crisis trigger with a sample message beforehand

Presentation Tips

1. **Start with the problem** — make the audience feel the gap before showing the solution
2. **Show, don't tell** — live demo > slides for technical features
3. **Emphasize the safety net** — crisis detection is the most impressive differentiator
4. **Use the PsychReport** — show the JSON output to demonstrate clinical depth
5. **Privacy angle** — "Everything runs locally" is a powerful statement
6. **Cultural touch** — the Bhagavad Gita remedies show thoughtfulness beyond just tech
7. **End with scale** — show the roadmap to demonstrate vision beyond the current version
8. **Keep it under 12 minutes** — leave time for Q&A
9. **Have backup screenshots** — in case the live demo fails

"PsyPredict doesn't just ask how you're feeling — it actually understands."
